Jiashen Wang

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EDUCATION

Carnegie Mellon University Pittsburgh, PA
M.S. in Applied Data Science, GPA: 3.83/4.0 Expected May 2026
B.S. in Statistics and Machine Learning, GPA: 3.86/4.0 Dec 2024

Courses: Applied Linear Models, Advanced Regression, Time Series, Experimental Design, Text Analysis, Statistical Programming, Data Visualization, Artificial Intelligence (AI), ML, Data Engineering, Algorithm & Data Structures

SKILLS

Technical: Python (pandas, numpy, Scikit-learn, PyTorch, Tensorflow), R, SQL, Azure, Spark, Git/Github, Excel, Java
Functional: Problem-Solving, Written and Verbal Communication Skills, Project Management, Teamwork, Collaboration

PROFESSIONAL EXPERIENCE

Data Scientist Jan 2016 – Present
Lucas Systems, in collaboration with CMU Pittsburgh, PA

- Manage a 3-member data science team as project lead under an Agile framework, coordinating with stakeholders on requirements, to deliver an end-to-end production solution from data ingestion to deployment for warehouse planning
- Develop and tune machine learning models (**Neural Networks, XGBoost**) on 116M+ records, improving task time prediction accuracy by 26% over existing heuristic-based methods
- Performed model selection and validation using cross-validation, optimizing for accuracy and runtime efficiency to ensure scalable, production-ready deployment

Teaching Assistant: Methods for Statistics & Data Science Aug 2025 – Present
Carnegie Mellon University Pittsburgh, PA

- Instructed 400+ students in data science techniques, covering predictive modeling, unsupervised learning, and visualization**,** strengthening students' data analytics skills and ability to deliver data-driven projects

Data Scientist Intern – Business Intelligence May 2025 – Aug 2025
Alibaba Beijing, China

- Designed **A/B tests** (500k+ users) on a product redesign, uncovering substantial improvements in engagement and monetization (+81.7% content exposure, +10.6% LTV), translating insights into product rollout decisions
- Enabled data-driven decision-making by developing KPIs management pipelines with embedded statistical analysis using SQL ETL workflows, adopted by cross-functional teams for real-time monitoring of performance metrics
- Applied unsupervised learning methods (**K-Means**) on 500k+ users, producing actionable customer segments for user growth strategy and communicating findings to non-technical business partners

Data Science Consultant Jan 2024 – Oct 2024
CMU Department of Statistics & Data Science Pittsburgh, PA

- Improved ECG data quality by developing a threshold-based signal outlier detection pipeline in R, detecting 98% of outliers across 500k+ recordings, and enhancing downstream modeling reliability at scale
- Engineered predictive features from raw time series signals and applied dimensionality reduction (PCA) to uncover latent disease structure, supporting downstream arrhythmia modeling and interpretation
- Applied **Lasso**-regularized logistic regression to classify arrhythmia, uncovering novel findings (0.92 Recall) and clinically useful signal features associated with disease, with results presented at CMU's research conference

PROJECTS

Distributed Big Data Analysis: Flight Delay Prediction with Spark Spring 2026

- Architected a cloud-based data processing system using PySpark on **Azure Databricks** to clean, transform, and analyze 10M+ flight records, performing data manipulation and feature engineering in a distributed environment
- Built Spark ML models to predict flight delays on large-scale data, enabling scalable, production-level forecasting

Loan Default Risk Classification using SBA Data Spring 2026

- Built an end-to-end data cleaning and EDA pipeline in Python (pandas) for a large financial dataset of 900K SBA loans
- Performed model selection and hyperparameter tuning via cross-validation across multiple ML algorithms (Random Forest, AdaBoost, XGBoost), achieving 93% accuracy, mirroring real-world credit risk workflows

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PROJECTS (Continued)

F1 Driver Speech Natural Language Processing (NLP) Fall 2025

- Engineered a web scraping and preprocessing framework for data collection of 1M+ tokens of unstructured data
- Applied hierarchical clustering and PCA to extract linguistic patterns and identify groups of drivers with similar communication styles, revealing interpretable differences in linguistic behavior

Automated Database and Dashboard Development for Retail Startup Spring 2025

- Designed a normalized **relational database** in **PostgreSQL** to centralize operational metrics of an Amazon store
- Developed an interactive dashboard (R Shiny) with automated ETL pipelines and dynamic visualizations to analyze 8 months of sales data (15k+ transactions), providing business insights and improving ad efficiency by 36%

Quantitative Time Series Modeling of Commodity Prices Spring 2024

- Modeled 60+ years of tea price data using a seasonal VAR(12) model to quantify cross-market lead–lag effects and uncover temporal dependencies across regional markets, providing insights into commodity market dynamics
- Evaluated model via out-of-sample **backtesting** and **volatility analysis**, identifying drivers to inform trading strategies

Automated Database and Dashboard Development for Retail Startup Spring 2025

- Designed a normalized **relational database** in **PostgreSQL** to centralize operational metrics of an Amazon store
- Developed an interactive dashboard (R Shiny) with automated ETL pipelines and dynamic **visualizations** to analyze **8 months of sales data (15k+ transactions)**, providing business insights and improving ad efficiency by 36%

- **Collaborated with stakeholders** through a structured design cycle to understand business requirements and ensure user experience, translating complex business data into actionable, data-driven insights

- Designed KPI dashboard with embedded statistical tests (SQL ETL), adopted by teams to make data-driven decisions
- Enabled data-driven decision-making by developing KPIs management pipelines with embedded statistical tests using SQL ETL workflows, adopted by cross-functional teams for real-time monitoring of performance metrics